Material Safety Data Sheet
#405 V.I. PLUS ENGINE OIL 15W/40

Issued: 8/22/95 Revised: 6/3/08

SECTION 1: CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

Frontier Performance Lubricants, Inc.
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PRODUCT NAME:	**FRONTIER LUBRICANT CODE 405 V.I. PLUS ENGINE OIL 15W/40**
CHEMICAL NAME:	**PETROLEUM LUBRICATING OIL**
CAS #:	**MIXTURE**
PRODUCT CATEGORY:	**PETROLEUM LUBRICANT**
MSDS CODE:	**405**
DATE ISSUED:	**8/22/95**
REPLACES ISSUE:	**FIRST ISSUE**

SECTION 2: COMPOSITION / INFORMATION ON INGREDIENTS

PRODUCT	**CAS NO. OF COMPONENTS**	
LUBRICATING BASE STOCK(S)	**MAY CONTAIN ONE OR MORE OF THE FOLLOWING:** CAS# 64742-54-4, 64742-65-0, 64742-58-1	>55%
PROPRIETARY ADDITIVES < 45%	MIXTURE	

PLEASE NOTE THAT THE CHEMICAL IDENTITY OF SOME OR ALL OF THE ABOVE HAZARDOUS INGREDIENTS IS CONFIDENTIAL INFORMATION AND MAY BE WITHHELD AS PERMITTED BY 29CFR 1910.1200 AND VARIOUS STATE RIGHT TO KNOW LAWS.

HAZARDOUS MATERIALS IDENTIFICATION SYSTEM (HMIS):

HEALTH	FLAMMABILITY	REACTIVITY
1	2	0

EXPOSURE LIMITS:

OIL MIST: ACGIH: TLV/TWA - 5 MG/M3, TLV/STEL - 10MG/M3
OSHA: PEL/TWA - 5 MG/M3, PEL/CEILING - NONE

SECTION 3: HAZARD IDENTIFICATION

VARIABLE AMONG INDIVIDUALS:
HEALTH STUDIES HAVE SHOWN THAT MANY PETROLEUM HYDROCARBONS AND SYNTHETIC LUBRICANTS POSE POTENTIAL HUMAN HEALTH RISKS WHICH MAY VARY FROM PERSON TO PERSON. AS A PRECAUTION, EXPOSURE TO LIQUIDS, VAPORS, MISTS OR FUMES SHOULD BE MINIMIZED.

EFFECTS OF OVEREXPOSURE (SIGNS AND SYMPTOMS OF EXPOSURE):
SKIN:

THIS MATERIAL MAY CAUSE SKIN IRRITATION AND/OR ALLERGIC SKIN REACTION. SYMPTOMS INCLUDE REDNESS, EDEMA, DRYING, DEFATTING AND CRACKING OF THE SKIN. PROLONGED AND REPEATED CONTACT MAY LEAD TO VARIOUS SKIN DISORDERS SUCH AS

DERMATITIS, OIL ACNE, OR FOLLICULITIS. PERSONS WITH PRE-EXISTING SKIN DISORDERS MAY BE MORE SUSCEPTIBLE TO THE EFFECTS OF THIS MATERIAL. BASED ON DATA FROM COMPONENTS OR SIMILAR MATERIALS.

EYE:
ONE COMPONENT OF THIS PRODUCT CONTAINS A SEVERE EYE IRRITANT.BASED ON DATA FROM COMPONENTS OR SIMILAR MATERIALS.

INGESTION:
MAY CAUSE IRRITATION OF THE DIGESTIVE TRACT, NAUSEA AND VOMITING. ONE COMPONENT (<1%) IS MODERATELY TOXIC AND MAY BE HARMFUL. ASPIRATION INTO THE LUNGS DURING INGESTION OR VOMITING MAY CAUSE MILD TO SEVERE PULMONARY INJURY AND POSSIBLE DEATH. BASED ON DATA FROM COMPONENTS OR SIMILAR MATERIALS.

RESPIRATORY: VAPORS OR MISTS MAY CAUSE IRRITATION OF THE NOSE AND THROAT, HEADACHE NAUSEA, AND DROWSINESS. HIGH CONCENTRATIONS MAY CAUSE HEADACHES, DIZZINESS, NAUSEA, STUPOR, AND OTHER CENTRAL NERVOUS SYSTEM EFFECTS LEADING TO VISUAL IMPAIRMENT, DIFFICULTY BREATHING AND CONVULSIONS. BASED ON DATA FROM COMPONENTS OR SIMILAR MATERIALS.

CHRONIC OVEREXPOSURE:
OVEREXPOSURE TO ONE OF THE COMPONENTS OF THIS MATERIAL HAS BEEN SUGGESTED AS A CAUSE OF THE FOLLOWING EFFECTS IN LABORATORY ANIMALS, AND MAY AGGRAVATE PRE-EXISTING DISORDERS OF THESE ORGANS IN HUMANS: MILD REVERSIBLE LIVER EFFECTS.

OTHER REMARKS:
WHEN OVERHEATED, PRODUCT MAY RELEASE HYDROGEN SULFIDE (H2S) GAS. H2S CONCENTRATIONS ABOVE PERMISSIBLE CONCENTRATIONS CAN CAUSE IRRITATION OF THE EYES AND RESPIRATORY TRACT, HEADACHE, DIZZINESS, NAUSEA, VOMITING. DIARRHEA AND PULMONARY EDEMA. AT CONCENTRATIONS ABOVE 300 PPM, RESPIRATORY PARALYSIS, CAUSING UNCONSCIOUSNESS AND DEATH, CAN OCCUR.

SECTION 4: FIRST AID MEASURES

EYE CONTACT:
IMMEDIATELY FLUSH EYES WITH LARGE AMOUNTS OF WATER FOR AT LEAST 15 MINUTES WHILE HOLDING EYELIDS OPEN. GET MEDICAL ATTENTION IF IRRITATION PERSISTS OR DEVELOPS.

SKIN:
IN CASE OF SKIN CONTACT, REMOVE ANY CONTAMINATED CLOTHING AND WIPE EXCESS OFF. WASH SKIN WITH SOAP AND WATER OR A WATERLESS HAND CLEANER FOLLOWED BY SOAP AND WATER. LAUNDER OR DRY-CLEAN CLOTHING BEFORE REUSE AND DISCARD SHOES AND OTHER LEATHER ARTICLES SATURATED WITH THE MATERIAL. GET MEDICAL ATTENTION IF IRRITATION PERSISTS OR DEVELOPS.

INHALATION:
REMOVE EXPOSED PERSON TO FRESH AIR IF ADVERSE EFFECTS ARE OBSERVED. GET MEDICAL ATTENTION IF BREATHING BECOMES DIFFICULT OR RESPIRATORY IRRITATION PERSISTS. IF BREATHING IS LABORED, ADMINISTER OXYGEN. IF BREATHING HAS STOPPED, APPLY ARTIFICIAL RESPIRATION.

INGESTION:
DO NOT INDUCE VOMITING, BECAUSE MATERIAL CAN ENTER THE LUNGS AND CAUSE SEVERE DAMAGE. DRINK 4 TO 8 OUNCES OF WATER AND SEEK IMMEDIATE MEDICAL ATTENTION. IF VOMITING OCCURS, OR IF COMATOSE OR LETHARGIC, PLACE ON THE LEFT SIDE WITH THE HEAD DOWN. SEEK IMMEDIATE MEDICAL ATTENTION.

SECTION 5: FIRE FIGHTING MEASURES

FLASH POINT:	150 DEG. F TCC (MIN)
UPPER FLAMMABILITY LIMIT:	NOT DETERMINED
LOWER FLAMMABILITY LIMIT:	NOT DETERMINED

EXTINGUISHING MEDIA AND FIRE FIGHTING PROCEDURES:
CARBON DIOXIDE, DRY CHEMICAL, FOAM OR WATER FOG. WATER SPRAY CAN BE USED TO COOL AND PROTECT CONTAINERS EXPOSED TO HEAT AND FLAME. PRODUCT WILL FLOAT ON SURFACE OF WATER.

SPECIAL FIREFIGHTING PROCEDURES:
RECOMMEND WEARING POSITIVE PRESSURE SELF-CONTAINED BREATHING APPARATUS. DO NOT ENTER CONFINED FIRE-SPACE WITHOUT FULL BUNKER GEAR (HELMET WITH FACE SHIELD, BUNKER COATS, GLOVES AND RUBBER BOOTS). TOXIC FUMES, GASES OR VAPORS MAY EVOLVE ON BURNING.

UNUSUAL FIRE & EXPLOSIVE HAZARDS:
HYDROGEN SULFIDE (H2S) MAY BE RELEASED IF OVERHEATED. HEATING ABOVE THE FLASH POINT MAY RELEASE FLAMMABLE VAPORS WHICH CAN BURN IN OPEN OR BE EXPLOSIVE IN CONFINED SPACES IF EXPOSED TO IGNITION. VAPORS MAY BE HEAVIER THAN AIR AND MAY TRAVEL ALONG THE GROUND TO A DISTANT IGNITION SOURCE AND FLASH BACK. IF EXPOSED TO IGNITION. MOVE CONTAINER FROM FIRE AREA IF YOU CAN DO SO WITHOUT RISK. APPLY COOLING WATER TO SIDES OF CONTAINERS THAT ARE EXPOSED TO FLAMES UNTIL WELL AFTER FIRE IS OUT. HEATED CONTAINERS MAY RUPTURE, EXPLODE OR BE THROWN INTO THE AIR.

DECOMPOSITION PRODUCTS UNDER FIRE CONDITIONS:
FUMES, SMOKE, CARBON MONOXIDE, ALDEHYDES HYDROGEN SULFIDE AND ALKYL MERCAPTANS AND SULFIDES AND OTHER PRODUCTS OF INCOMPLETE COMBUSTION. UNDER COMBUSTION CONDITIONS, OXIDES OF THE FOLLOWING ELEMENTS WILL BE FORMED: CALCIUM, MAGNESIUM, SULFUR.

SECTION 6: ACCIDENTAL RELEASE MEASURES

STEPS TO BE TAKEN IN CASE MATERIAL IS RELEASED OR SPILLED:
REMOVE ALL SOURCES OF IGNITION. PERSONNEL PROTECTIVE EQUIPMENT MUST BE WORN(SEE SECTION VIII). VENTILATE SPILL AREA IF IN CONFINED OR POORLY VENTILATED AREA. PICK UP FREE LIQUID USING EXPLOSION PROOF OR HAND PUMP FOR RECYCLE AND/OR DISPOSAL IF IT CAN BE DONE SAFELY. RESIDUAL MATERIAL CAN BE ABSORBED ON INERT MATERIAL. MINIMIZE SKIN CONTACT. KEEP PRODUCT OUT OF SEWERS AND WATERCOURSES BY DIKING OR IMPOUNDING. AVOID CONTACT WITH SKIN, EYES OR CLOTHING.

WASTE DISPOSAL:
WASTE DISPOSAL SHOULD BE IN COMPLIANCE WITH ALL FEDERAL, STATE, AND LOCAL LAWS.

SECTION 7: HANDLING AND STORAGE

STORE IN A COOL DRY LOCATION. KEEP AWAY FROM POTENTIAL SOURCES OF IGNITION. KEEP AWAY FROM INCOMPATIBLE MATERIALS. OPEN CONTAINER IN A WELL VENTILATED AREA. AVOID GENERATING MISTS WHEN HANDLING. AVOID BREATHING VAPORS, OR PROLONGED OR REPEATED SKIN CONTACT. WASH THROUGHLY AFTER HANDLING. REMOVE ANY CONTAMINATED CLOTHING AND LAUNDER BEFORE REUSE. MINIMUM FEASIBLE HANDLING TEMPERATURES SHOULD BE MAINTAINED. PERIODS OF EXPOSURE TO HIGH TEMPERATURES SHOULD BE MINIMIZED. STATIC ELECTRICITY MAY ACCUMULATE AND CREATE A FIRE HAZARD. GROUND ALL CONTAINERS 5 GALLONS AND LARGER.

"EMPTY" CONTAINER WARNING:
"EMPTY" CONTAINERS RETAIN RESIDUE (LIQUID AND/OR VAPOR) AND CAN BE DANGEROUS. **DO NOT PRESSURIZE, CUT, WELD, BRAZE SOLDER, DRILL, GRIND, OR EXPOSE SUCH CONTAINERS TO HEAT, FLAME, SPARKS, STATIC ELECTRICITY, OR OTHER SOURCES**

OF IGNITION; THEY MAY EXPLODE AND CAUSE INJURY OR DEATH. DO NOT ATTEMPT TO CLEAN SINCE RESIDUE IS DIFFICULT TO REMOVE. "EMPTY" DRUMS SHOULD BE COMPLETELY DRAINED, PROPERLY BUNGED AND PROMPTLY RETURNED TO A DRUM RECONDITIONER. ALL OTHER CONTAINERS SHOULD BE DISPOSED OF IN AN ENVIRONMENTALLY SAFE MANNER AND IN ACCORDANCE WITH GOVERNMENTAL REGULATIONS.

SECTION 8: EXPOSURE CONTROLS - PERSONAL PROTECTION

VENTILATION:
PROVIDE EXPLOSION PROOF VENTILATION SUFFICIENT TO PREVENT EXCEEDING RECOMMENDED EXPOSURE LIMIT. USE LOCAL EXHAUST TO CAPTURE VAPOR, MISTS OR FUMES, IF NECESSARY. NO SMOKING, FLAME OR OTHER IGNITION SOURCES.

RESPIRATORY PROTECTION:
UNDER NORMAL USE CONDITIONS, RESPIRATOR IS NOT USUALLY REQUIRED. USE NIOSH/MSHA APPROVED AIR-PURIFYING RESPIRATOR FOR ORGANIC VAPORS AND PARTICULATES IF THE RECOMMENDED EXPOSURE LIMIT IS EXCEEDED. USE SELF-CONTAINED BREATHING APPARATUS FOR ENTRY INTO CONFINED SPACE, FOR OTHER POORLY VENTILATED AREAS AND FOR LARGE SPILL CLEAN-UP SITES.

PROTECTIVE GLOVES:
USE CHEMICAL RESISTANT GLOVES TO AVOID PROLONGED OR REPEATED SKIN CONTACT. NITRILE GLOVES ARE RECOMMENDED.

EYE PROTECTION:
USE SPLASH GOGGLES OR FACE SHIELD WHEN EYE CONTACT MAY OCCUR.

OTHER PROTECTIVE EQUIPMENT:
USE CHEMICAL RESISTANT BOOTS AND APRONS OR OTHER IMPERVIOUS CLOTHING, IF NEEDED, TO AVOID CONTAMINATING REGULAR CLOTHING. LONG SLEEVE SHIRT IS RECOMMENDED.

WORK PRACTICES:
KEEP CONTAINERS CLOSED WHEN NOT IN USE. DO NOT STORE NEAR HEAT, SPARKS, FLAME OR STRONG OXIDANTS.

PERSONAL HYGIENE:
MINIMIZE BREATHING VAPOR, MIST OR FUMES. AVOID PROLONGED OR REPEATED CONTACT WITH SKIN. REMOVE CONTAMINATED CLOTHING; LAUNDER BEFORE REUSE. REMOVE CONTAMINATED SHOES AND THOROUGHLY CLEAN BEFORE RE-USE; DISCARD IF OIL-SOAKED. CLEANSE SKIN THOROUGHLY AFTER CONTACT, BEFORE BREAKS AND MEALS, AND AT END OF WORK PERIOD.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

SPECIFIC GRAVITY: 0.87 - 0.89 @ 15.6C	**VAPOR PRESSURE:** NOT DETERMINED
% VOLATILE BY VOLUME: NOT DETERMINED	**VAPOR DENSITY (AIR=1)** HEAVIER THAN AIR
SOLUBILITY IN WATER: NOT DETERMINED	**EVAPORATION RATE (n-BUTYL ACETATE =1)** NOT DETERMINED

SECTION 10: STABILITY AND REACTIVITY

STABILITY:	STABLE UNDER NORMAL CONDITIONS OF STORAGE AND HANDLING.
HAZARDOUS POLYMERIZATION:	WILL NOT OCCUR
AVOID CONTACT WITH:	ACIDS, OXIDIZING AGENTS, HALOGENS AND HALOGENATED COMPOUNDS.

SECTION 11: ECOLOGICAL INFORMATION

REPORTABLE QUANTITY (RQ), EPA REGULATION 40 CFR 302 (CERCLA SECTION 102):
NO RQ FOR PRODUCT KNOWN TO BE GREATER THAN 1% OR 01% (CARCINOGEN).

SARA TITLE III:
SECTION 313 REPORTABLE INGREDIENTS:
THIS PRODUCT CONTAINS THE FOLLOWING SUBSTANCES SUBJECT TO THE REPORTING REQUIREMENTS OF SARA III SECTION 313 1986 AND 40 CFR PART 372: FROM 0.2 TO 2.0%, ZINC COMPOUNDS; CONTAINING 0.02 TO 0.2 AS Zn.

SECTION 311/312:
IMMEDIATE, DELAYED HEALTH

CALIF. PROP. 65:
THIS PRODUCT MAY CONTAIN CHEMICALS KNOWN TO THE STATE OF CALIFORNIA TO CAUSE CANCER AND/OR BIRTH DEFECTS.

SECTION 12: TRANSPORTATION INFORMATION

DOT PROPER SHIPPING NAME FOR NON-BULK:	NOT APPLICABLE
DOT HAZARD CLASS:	NOT REGULATED
DOT UN NUMBER:	NOT APPLICABLE

SECTION 13: OTHER INFORMATION

THE INFORMATION AND RECOMMENDATIONS HEREIN ARE, TO THE BEST OF OUR KNOWLEDGE AND BELIEF, ACCURATE AND RELIABLE AS OF THE DATE ISSUED. FRONTIER DOES NOT WARRANT OR GUARANTEE THEIR ACCURACY OR RELIABILITY, AND SHALL NOT BE LIABLE FOR ANY LOSS OR DAMAGE ARISING OUT OF THE USE THEREOF.

THE INFORMATION AND RECOMMENDATIONS ARE OFFERED FOR THE USER'S CONSIDERATION AND EXAMINATION, AND IT IS THE USER'S RESPONSIBILITY TO SATISFY ITSELF THAT THEY ARE SUITABLE AND COMPLETE FOR ITS PARTICULAR USE. IF BUYER REPACKAGES THIS PRODUCT, LEGAL COUNSEL SHOULD BE CONSULTED TO INSURE PROPER HEALTH, SAFETY AND OTHER NECESSARY INFORMATION IS INCLUDED ON THE CONTAINER.